

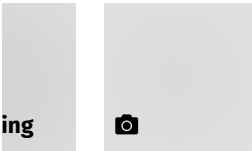
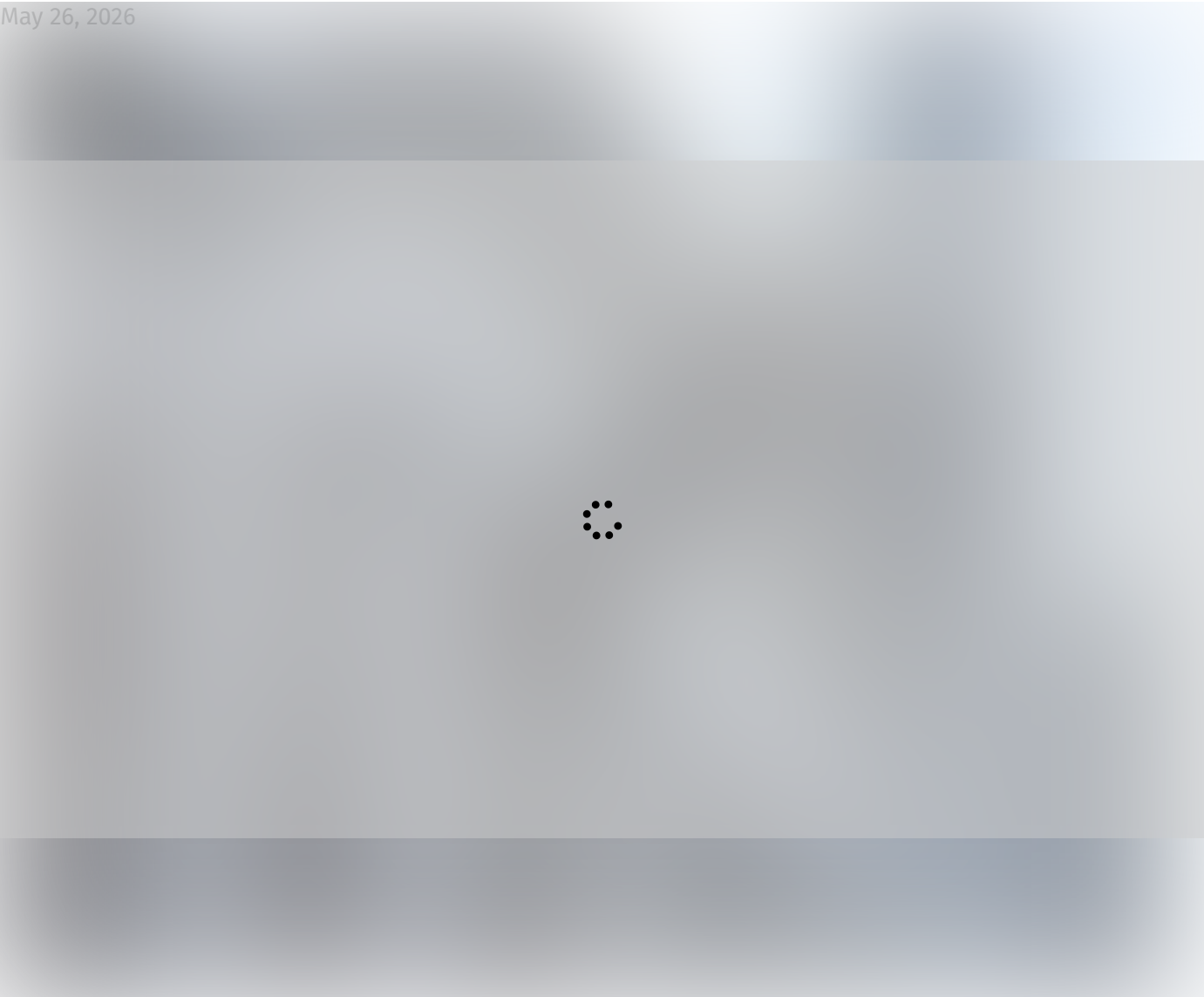
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FEATURED

Flock cameras: How they work and why some are concerned

Sydney Dawes and Cornelius Frolik - Staff Writers

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Cornelius Frolik / Staff

A Flock license plate reader on Xenia Avenue in East Dayton. Cornelius Frolik / Staff

Most motorists probably do not even notice them, but automated license plate readers are everywhere, constantly recording the plates and other visual information of passing cars in communities all across southwest Ohio.

Officers with local police departments routinely search the data the devices collect for investigative leads and other purposes. Police also share their databases with outside agencies.

A recent Dayton Daily News investigation found that local jurisdictions allow outside law enforcement agencies to scan their ALPR databases thousands of times every day for purposes that are not always clear, since officers can provide broad, vague and unconfirmed reasons for their searches.

The city of Dayton suspended use of its Flock cameras — a type of ALPR that generates data that can be accessed by departments all over the country — after realizing its data was being scanned for immigration reasons in violation of city policy. The Dayton Daily News investigation found other area police departments also allow their data to be accessed for immigration purposes.

How police use and share ALPR data is a hot-button issue. Privacy advocacy groups say ALPRs are a powerful mass surveillance tool that has a grave potential for misuse, since the information they collect can be accessed with little to no accountability. Police officials claim they provide valuable evidence in cases that otherwise would go unsolved. This story explains how the technology works so community members can draw their own conclusions.

What are ALPRs?

ALPR cameras usually are mounted to poles or traffic signals and capture images of passing vehicles' license plate numbers and other information. Police agencies also use dashboard-mounted ALPRs in police cruisers.

Flock is a leading maker of ALPR technology, but there are other manufacturers, like Axon. Not all ALPR systems allow captured data to be stored and shared with other departments to the same extent.

DAYTON DAILY NEWS INVESTIGATES

Automated license plate readers



The city of Dayton recently suspended its use of fixed-point automatic license plate readers after disclosing camera data was being accessed by outside agencies for immigration-related searches.

We obtained data from other area police departments to reveal how ALPR data is shared, and for what reasons.

» Area police license plate reader data routinely accessed for immigration investigations, records show

» Privacy advocates: Legislation, guardrails needed for police license plate readers

» How Kettering, other area police departments share license plate reader data

Flock cameras capture the make, model and color of passing vehicles, as well as other visual details, like damage or alterations, such as broken taillights and aftermarket rims or wheels. Cameras can grab images of vehicles traveling as fast as 100 mph, day and night, up to 75 feet away.

ALPR cameras typically send real-time alerts when license plate numbers match a reported stolen vehicle or a car being sought by law enforcement. But the technology can do a lot more than that.

Organizational audit records for Flock systems for police in Kettering, Miami Twp. and Tipp City obtained by this news outlet show officers have performed ALPR database searches and look-ups for wanted persons, welfare checks, drug investigations, criminal motor vehicle offenses, hit-and-runs and crashes, intoxicated driving violations, thefts, protection order cases, traffic infractions, financial crimes and many other purposes.

Police officials says ALPRs are a vital tool that greatly improves public safety by alerting officers in real time to vehicles associated with criminal activity. Officials say plate readers have helped police identify, track and locate getaway vehicles. Unlike traditional video cameras, Flock cameras do not require police to go through hours of security video to find what they are looking for.

“As with any investigative tool, Flock searches are used to assist officers and detectives in locating vehicles, identifying movement patterns, corroborating investigative information and generating leads during criminal investigations,” said Miamisburg police Chief Mike Brem.

“The system is one tool among many and is most effective when combined with traditional investigative work, officer observations, witness statements and other evidence.”

Flock camera technology has helped police in Tipp City make more than 100 arrests for serious criminal offenses, and the devices have assisted officers who were investigating stolen vehicles, theft offenses, wanted suspects and missing persons, said Tipp City Manager Eric Mack.

But while these cameras are supposed to capture the “fingerprint” of vehicles, they also may collect a much wider range of information, said Phu Phung, a professor at the University of Dayton.

According to Phung, systems like ALPR networks typically use artificial intelligence or other advanced software to pull data from images: The color, make and model of the vehicles detected, as well as other data linked to time and location.



University of Dayton Professor Phu Phung talks about his cybersecurity research and issues organizations should consider when implementing data-sharing technology. SYDNEY DAWES / STAFF

But Phung said there's no telling what other details are captured within an image recorded by the cameras: People inside a vehicle, the surrounding street or other information that is framed in each snapshot of time. Privacy advocacy groups say cameras record details like bumper stickers and defects — cracks in windshields or other identifiable damage — that could be analyzed and stored.

"There is no technical way to verify that that is what they capture," Phung said. "As a researcher, I'm a little bit skeptical about that."

ALPRs can be used for legitimate law enforcement purposes, but police likely are tempted to use the technology and data they collect for all kinds of reasons, said Gary Daniels, legislative director for the ACLU of Ohio.

“This comes down to the fundamental question of, How much do you trust your government?” Daniels said.

Daniels said police departments’ use of ALPRs are largely unregulated in Ohio. He said few places have strong policies about under what circumstances the data can be searched and shared. He said too often law enforcement has carte blanche access to massive amounts of data with no oversight.

Phung said each Flock camera operates like a small computer among a broader system of computers that are connected via the cloud. These small computers use motion detection to operate and cloud-based storage to preserve data, and they are often powered by solar panels.

Data collected through the ALPR camera system’s pre-processing stages is then uploaded to a cloud-based server. This data is encrypted to limit its access, but policies around who can access it and when are up to the customers and technology providers. The same goes for how long the data is stored. Phung said some cities opt for a 30-day erasure system while others hold on to data not needed for criminal investigations much longer.

The Dayton Police Department and Dayton city manager’s office announced this month a halt to the city’s fixed-site automated license plate reader program. The pause was ordered after a review of the cameras revealed that a network-sharing feature had been enabled, allowing data to be accessed more broadly than what was authorized.

This announcement was met with outrage from multiple citizens, who called the policy violation a betrayal of public trust. City leaders, too, have called for the removal of the fixed-site cameras altogether or the review of all city data-sharing policies.

Phung’s research proposes a dynamic, context-aware approach to data access.

“I think the biggest challenge is that every city, if they want to deploy this system, they need to outsource this to a company, and probably a private company,” Phung said.

“We need to have some kind of a more formal way, technical way, to verify that the system followed, is there.”

In Dayton's announcement about the Flock incident, the Dayton Police Department also said a major within its force no longer works for the agency, promising to conduct an internal investigation into the matter.

Phung said audits of data-sharing systems do not prevent any breach to an organization's cybersecurity. Rather, they flag agencies to issues after the fact — "already too late," Phung said.

"I would like to say that if a human flaw causes the system flaw, that is a design flaw," Phung said.

Many local police agencies have denied or not responded to the Dayton Daily News' public record requests seeking network and internal Flock ALPR audits. Without this information, it is impossible to know the extent of data-sharing with outside agencies and the reasons police are giving to search the databases.

Agencies that denied this news outlet's records requests or failed to respond or provide the records include the city of Dayton; Huber Heights; Middletown; the Montgomery County Sheriff's Office; the Clark County Sheriff's Office; the Miami County Sheriff's Office; and the Butler County Sheriff's Office. Springboro and Miamisburg provided limited records. Centerville police said they do not have the requested records.

Explaining license plate reader history, access issues.

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